

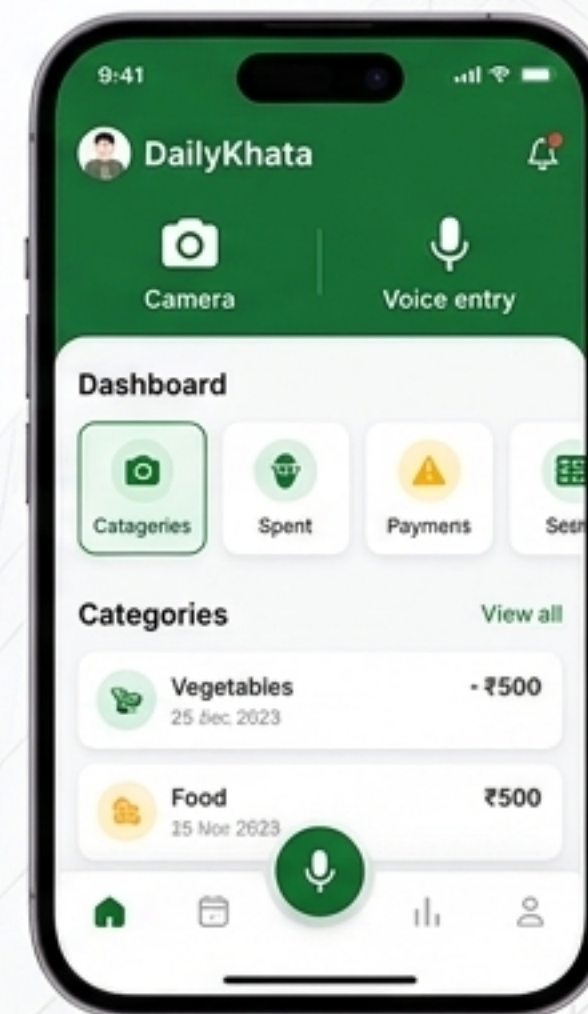
From Messy Reality to Structured Clarity: AI-Powered Expense Tracking

Team Patches (Saad Mughal, Khadijah Zahoor) | Milestone 02



Khata Green

*Manual entry is tedious.
Existing tools demand rigid categorization.
Local receipts fail standard OCR.*



Camera-to-AI

Snap any receipt
(printed/handwritten)
for auto-extraction.



Voice-First Entry

Say "Spent 500 on
vegetables" to log
instantly.



Live Spreadsheet UI

Power-user grid for
bulk edits.



Touch-to-Edit

Tap any AI-extracted
field to correct errors
instantly.

Systematically Uncovering the Friction in Financial Tracking

Study Goal: Understand how 18-50yo Pakistanis track expenses and why they abandon digital tools.
(30-45 min sessions).

Step 1

Intro & Consent

10-20 mins.
Anonymous.
Audio recorded.

Step 2

Warm-up

Who handles shopping? What tools do you use? What is the most frustrating part?

Step 3

The Shopping Trip

Walk us through your last grocery run. Printed or handwritten receipt? Where did you put it?

Step 4

Workflow Observation

Show us your system. How do you record 'milk and bread'? How do you categorize? Time spent per week?

Step 5

Friction & Abandonment

What stopped you from recording? How do you handle rickshaw cash? Why did you quit other apps?

Step 6

Concept Intro

If you could snap a photo or speak, how would it change your routine? Would you trust AI? What if it's unsure?

Step 7

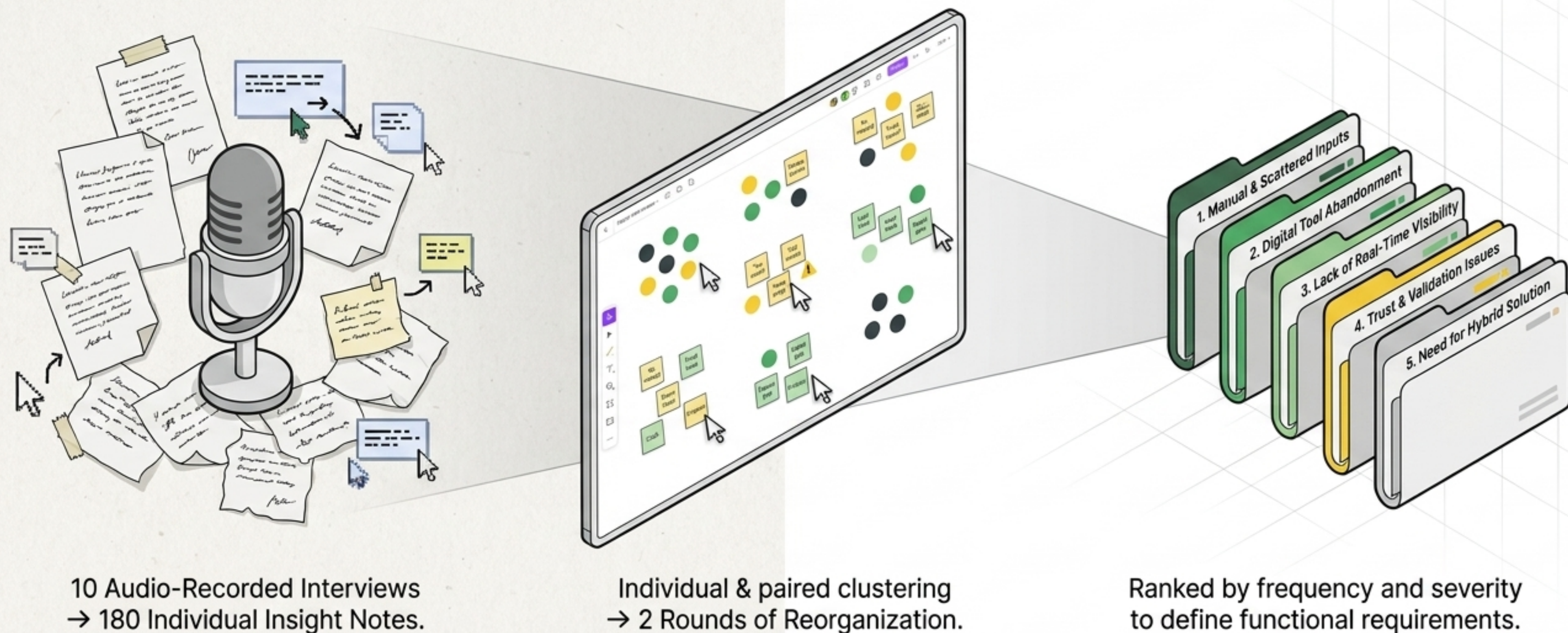
Conclusion

Wrap up & next steps.

Diverse Profiles Spanning Analogue Traditions and Digital Frustrations

Group A - Homemakers & Shopkeepers			
ID	Profile	Digital Literacy	Current Method
P1	Homemaker, 40s	Low digital literacy	Physical notebook (Khata)
P2	Homemaker, 30s	Mod digital literacy	Notebook + WhatsApp notes
P3	Head of household, 50s	Low digital literacy	Mental tracking + receipts in drawer
P4	Local shopkeeper, 35	Mod digital literacy	Paper ledger
P5	Homemaker, 28	Mod digital literacy	Combination of notes app + notebook
Group B - Students & Freelancers			
P6	Uni student, 21	High digital literacy	Abandoned app (Wallet); now spreadsheet
P7	Uni student, 22	High digital literacy	No consistent system
P8	Freelancer, 26	High digital literacy	Notion + manual receipt photos
P9	SME owner, 30	Mod digital literacy	WhatsApp messages to self
P10	Uni student, 23	Mod digital literacy	Abandoned Cashew app; now mental tracking

From 180 Raw Observations to 5 Core Themes



Our process prioritized direct observation of physical workflows over self-reported habits, exposing critical gaps in digital trust.

The Affinity Wall: Mapping the Reality of Pakistani Spending

Receipt Chaos (38 Notes)

Lost it on
the way
home

Can't
read the
number

Don't
bother
saving it

Category Mismatch (29 Notes)

No ration
category

Groceries
doesn't fit
my spending

I don't
think like
that

Informal Cash Gaps (31 Notes)

Rickshaw
has no
receipt

Vendor just
says the
amount

No way
to log this

Manual Entry Fatigue (42 Notes)

Takes too
long

Gave up
after a
week

Feels like
extra
homework

AI Trust / Verification (40 Notes)

What if
it reads
wrong

I need to
double-
check

Show me
what it
understood

Identifying the Friction: Why Current Practices Fail



Practice

Crumpled in purses, tossed in drawers.

Need

A system that digitizes data before the physical paper is lost.



Practice

Abandoning apps because typing every purchase feels like a 'second job.'

Need

Zero-typing data entry.

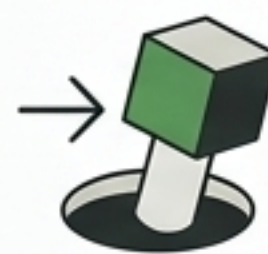


Practice

Universally skipping roadside stalls and rickshaws (30-40% of spending).

Need

A logging method for transactions with zero paper trail.



Practice

Thinking in 'occasions' (weekly bazar) rather than Western app categories.

Need

Flexible, locally-contextualized tagging.



Practice

Refusing to rely on automation if it can't handle mixed Urdu/English.

Need

Absolute transparency and easy error-correction.

Balancing Ambition with Reality: Requirements & Constraints

Functional Requirements (The Ambition)

-  - **Core:** Camera pipeline for handwritten/faded/mixed-language receipts. Voice input for informal cash. Inline verification step (touch-to-edit).
- 
- 
-  - **Secondary:** Flexible local categories. Zero-effort weekly/monthly plain-language summaries.
- 
-  - **Optional/Pro:** Offline capability for low-connectivity areas.



Identified Constraints (The Reality)

- **Technical:** Must run smoothly on low-end Android 9+ devices with 2GB RAM. Background sync for offline logging. 

- **Usability:** Zero mandatory account creation at launch. 'Misc' must always be a valid, unforced bucket. 

- **Financial:** No subscription paywall on core features (highly price-sensitive demographic). 

The Strategic Pivot: From OCR-Only to Multimodal Trust

Initial Assumption



We believed generic OCR would solve the problem.

Research revealed OCR fails on mixed Urdu/English, and users fundamentally distrust AI they cannot verify.

Updated Project Brief



- **Refined Target:** Low-literacy homemakers (need zero-friction) AND moderate-literacy students (need data density).
- **Updated Solution: DailyKhata.** Three equal input modes (Camera, Voice, Manual) united by a mandatory inline edit-before-save verification.
- **UI Philosophy:** Minimal default (one tap to log) with a hidden 'Pro View' for power users. Functional onboarding in under 60 seconds.

Designing for Two Distinct Realities

Rukhsana Bibi (Ammi Jaan - 43)

 Lahore Homemaker | 40k/mo budget | Android (3GB RAM).

 **Current Tool**
Physical Khata notebook. Receipts in a drawer.

 **Digital Literacy**
 20%
Low - WhatsApp only

 **Tech Trust**
 10%
Distrustful of AI

 **Pain Point**
Faded receipts, forgets small items, apps feel "too English".

"Main bas ye jaanna chahti hoon ke hafte mein kitna gaya."

Zaid Malik (The Hustle Student - 22)

 Final-year CS student & Freelancer | Android (4GB RAM).

 **Current Tool**
Notion table (abandoned Cashew app).

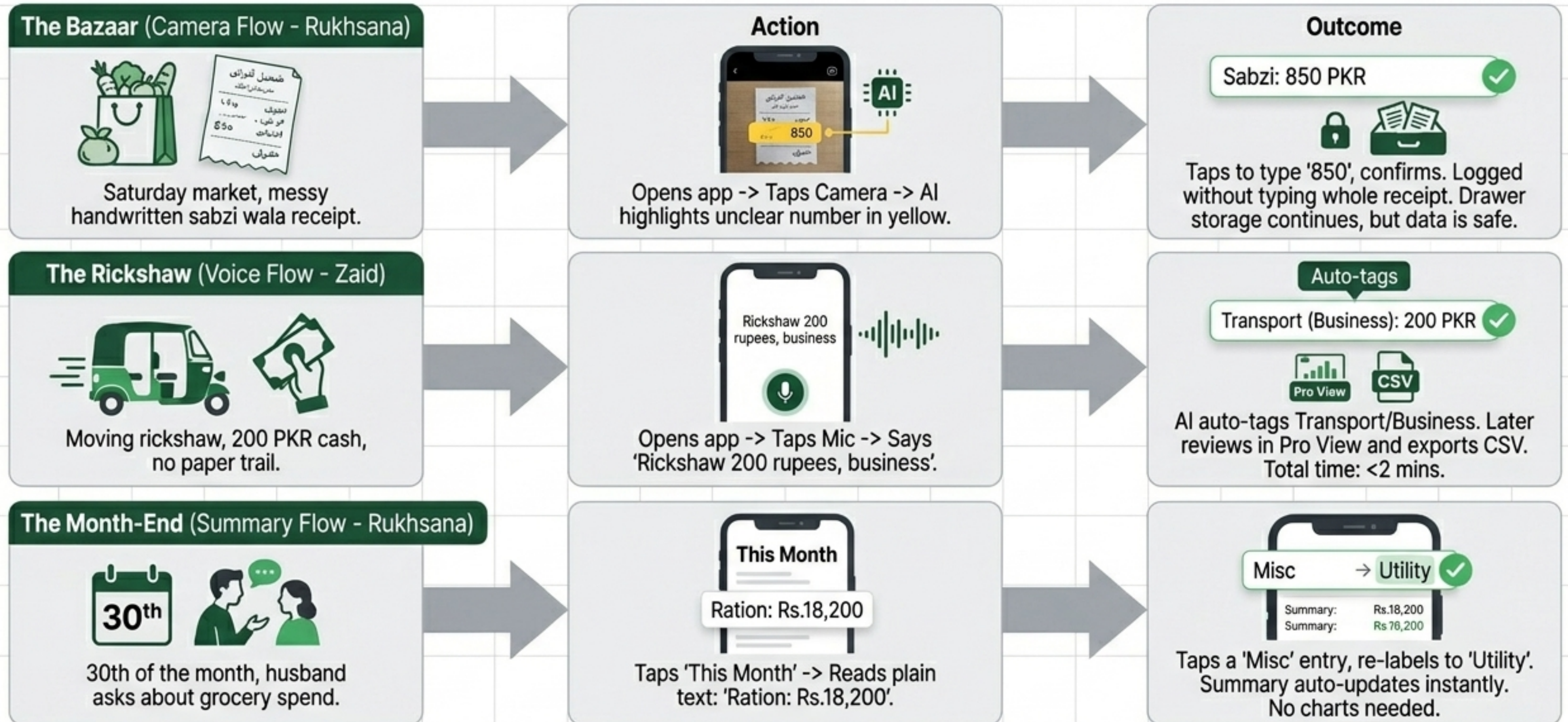
 **Digital Literacy**
 95%
High

 **Tech Trust**
 80%
Wants transparency and Pro tools

 **Pain Point**
Categorization takes too long; mixed PKR/foreign gig income.

"I want to spend 5 seconds logging, not 5 minutes."

Contextual Scenarios: DailyKhata in the Real World

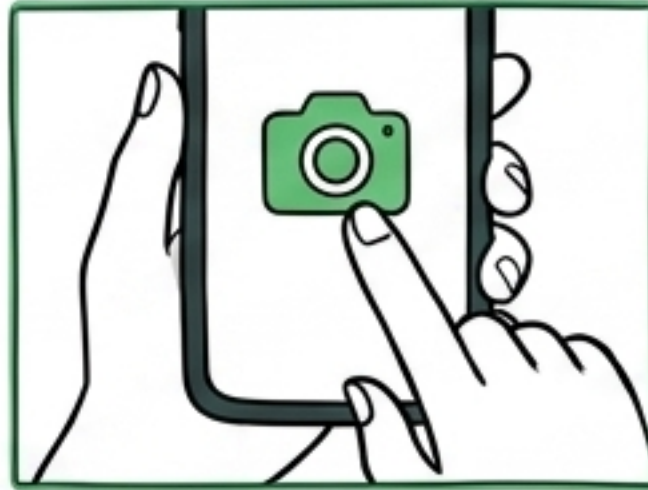


Storyboards: Frictionless Multimodal Entry

Camera Scan with Verification



User returns with messy receipt.



One tap on home screen camera icon.



AI processes; unsure fields highlighted in bright yellow.



User taps yellow field, uses numpad to correct.



Taps 'Confirm'. Daily total updates. (<30 seconds).

Voice Logging for Informal Cash



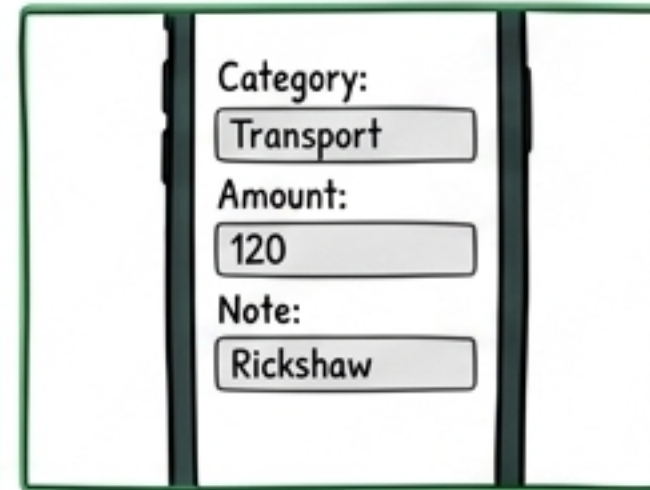
User pays rickshaw driver cash.



Taps giant microphone icon.



Speaks naturally:
"Spent 120 on rickshaw."



Screen shows parsed intent.

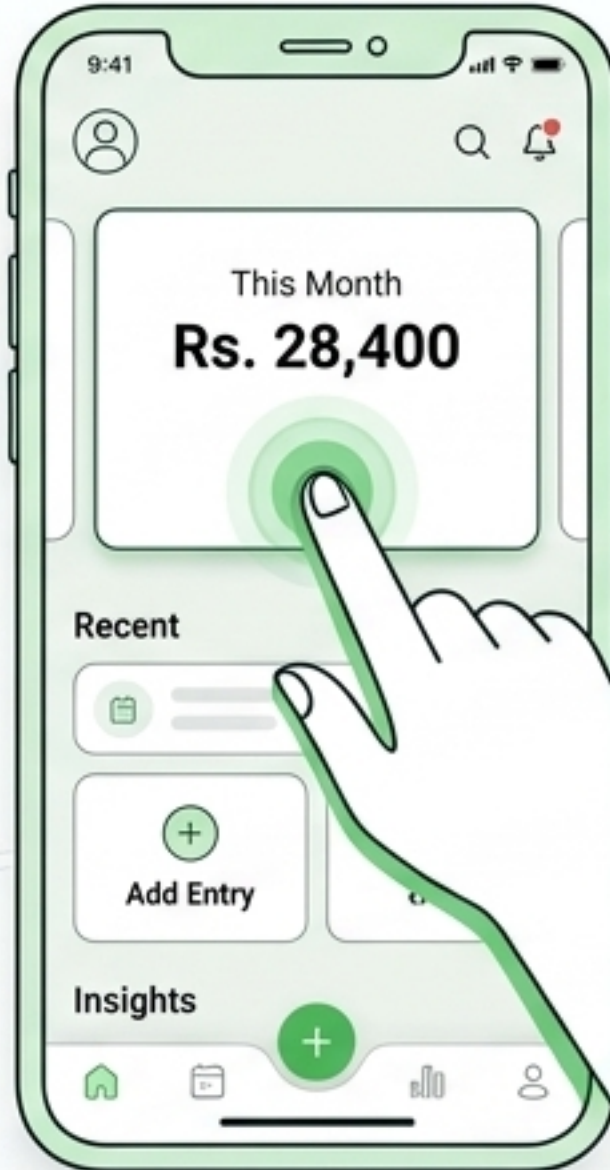


Taps 'Confirm'. Done.
(<15 seconds, zero typing).

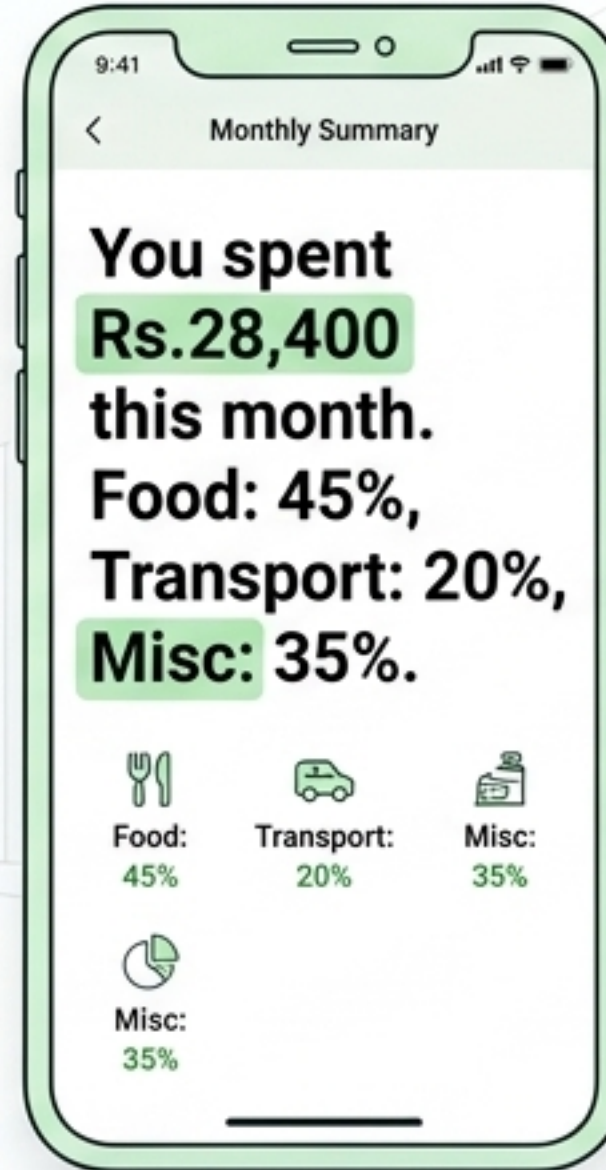
Storyboards: The Plain-Language Monthly Review



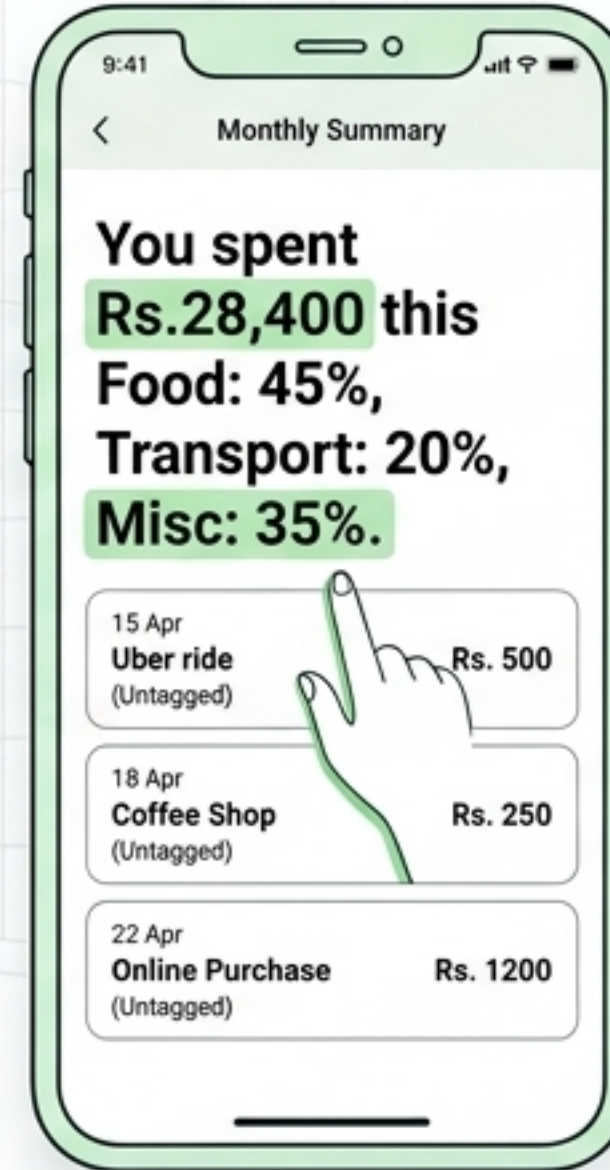
End of the month. User wants to understand total spending without complex charts.



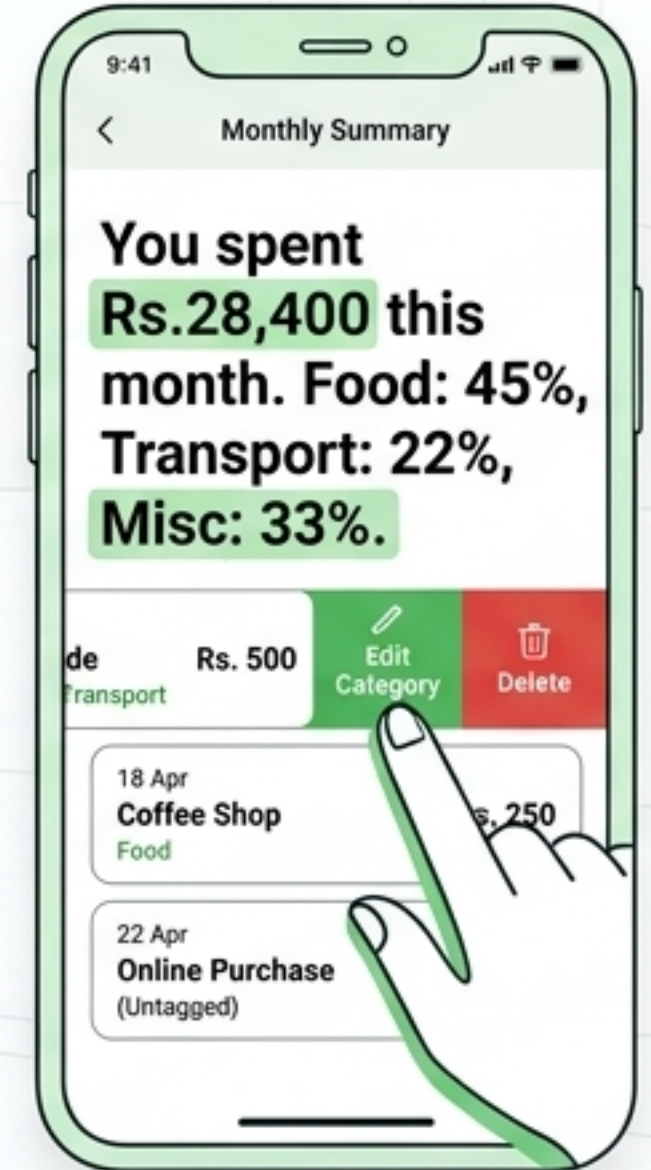
Single tap on the 'This Month' card on the home screen.



UI displays a plain-language sentence: "You spent Rs.28,400 this month. Food: 45%, Transport: 20%, Misc: 35%."



User taps the word 'Misc'. A list of individual, untagged transactions drops down.



User swipes to re-categorize two entries. The main summary sentence auto-updates seamlessly.

Mapping the Competitive Landscape

Market Filters (Selection Criteria)



- Target personal/household expenses.



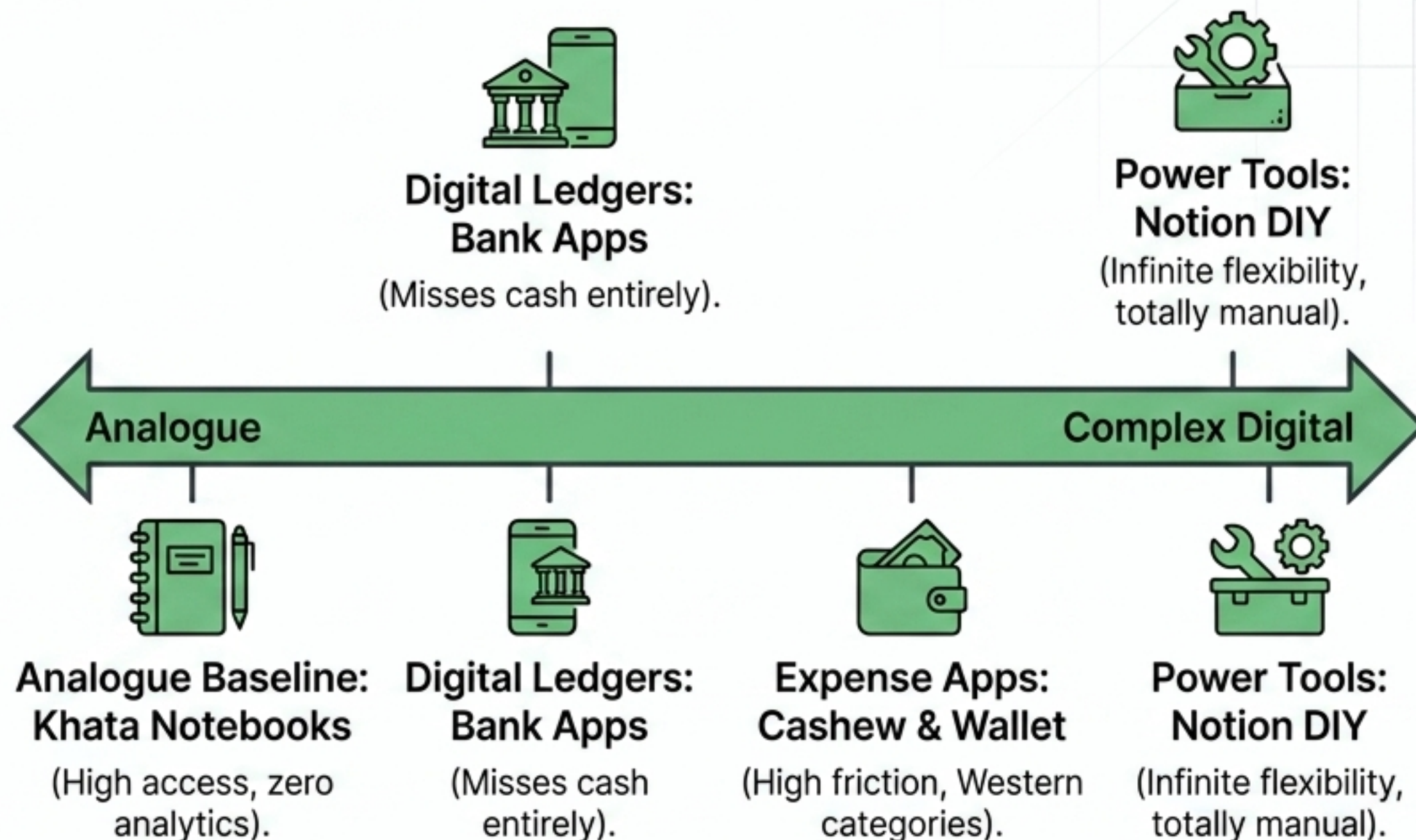
- Mobile application availability in Pakistan.



- Features OCR, Voice, or AI categorization.



- Evaluated via App Store UX reviews and direct testing.



Competitive Matrix: Why Existing Solutions Fail Pakistan

Features	Khata Notebook	Bank App	Cashew	Wallet	Notion	DailyKhata
Receipt Scanning	✗	✗	Basic OCR	Basic OCR	✗	AI (handwritten/mixed)
Voice Input	✗	✗	✗	✗	✗	✓
Informal Cash Logging	Manual	✗	Manual	Manual	Manual	One-tap voice
Urdu/Local Receipt Support	✓	✗	✗	✗	N/A	✓
Inline Verification	N/A	N/A	Auto-commit	Auto-commit	Manual	Edit before save
Local Categories	User-defined	N/A	Western	Western	User-defined	Local defaults
Offline Functionality	✓	✗	Partial	Partial	✓	✓
Onboarding Complexity	None	High	Medium	High	High	Minimal (<60s)

****Only DailyKhata supports Voice Input (NLP) and Urdu/Local Receipt Support. Wallet/Cashew force Auto-commit; DailyKhata uniquely uses Inline edit before save.**

Market Validation & Strategic Opportunities

Best Practices to Adopt

- ⚙️ **Automated Categorization:** Users hate thinking about categories (validation from Cashew/Wallet).
- 📷 **Camera Willingness:** 'Smart Receipt' features show users *want* to scan; the tech just needs to be better.
- 📄 **Raw Data Access:** Notion's popularity proves power users need CSV exports.

Opportunities for Differentiation

- ✍️ **The Handwriting Gap:** No competitor addresses mixed Urdu/English handwritten receipts.
- 🎤 **Voice as Primary:** Entirely absent in the market, yet perfect for un-receipted rickshaw rides.
- ✅ **The Trust Loop:** Inline editing replaces auto-commit, solving the primary AI trust deficit.
- 🌐 **Offline-First:** Uniquely designed for spotty connectivity zones.

DailyKhata bridges the gap between analogue reality and digital analytics.